

SEC P vs NP log 2026-05-11

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-11 05:00 UTC

SEC P vs NP — daily log 2026-05-11

15 cycles recorded

Homotopy Group Rank Inverse Proportional to Disjointness Communication Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic Topology (Homotopy Theory) $\times$ Disjointness Communication Complexity
- **Recorded:** 2026-05-11 00:01 UTC
- **Entry ID:** c0f03be34ae3

Statement

For any bipartite graph $G = (U \cup V, E)$ with $|U| = |V| = n$, the rank of the first homotopy group $\pi_1(\Gamma_G)$ of its communication matrix Γ_G is $\Theta(1/CC(G))$, where $CC(G)$ is the deterministic communication complexity of disjointness on G .

Rationale

Homotopy invariants capture global structural constraints of communication matrices, which may correlate with inherent complexity barriers. The fundamental group's rank reflects non-trivial loops in the matrix's topology, potentially mirroring the information-theoretic cost of coordination.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a1218782.py", line 86, in <module>
    result = run_trial(seed)
             ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a1218782.py", line 68, in run_trial
    M = clique_incidence_matrix(G)
        ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a1218782.py", line 37, in clique_incidence
    M[u][v+n] = 1
    ~~~~^
```

IndexError: list assignment index out of range

Judge reasoning

Test crashed with IndexError, preventing data collection. Support condition cannot be evaluated without successful trials. | next: Fix the clique_incidence_matrix implementation to handle index bounds properly

Diophantine Approximation Error Inversely Proportional to Communication Discrepancy in XOR-Functions

- **Verdict:** INCONCLUSIVE
- **Bridge:** Diophantine Approximation $\times$ Communication Complexity of XOR-Functions
- **Recorded:** 2026-05-11 00:27 UTC
- **Entry ID:** e98d1643eeaa

Statement

For an XOR-function $f: \{0,1\}^n \times \{0,1\}^n \rightarrow \{0,1\}$, let $D(f)$ denote its communication discrepancy and $\varepsilon(f)$ denote the minimal approximation error of its Fourier coefficients by rational numbers with denominator $\leq n^2$. Then $D(f) \geq \Omega(1/\varepsilon(f) \cdot \log n)$.

Rationale

Diophantine approximation quantifies how well real numbers can be approximated by rationals, while discrepancy measures imbalance in communication matrices. Linking them could reveal structural constraints on XOR-functions' Fourier spectra, potentially separating complexity classes via approximation-theoretic barriers.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 1.91s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9b764fd6.py", line 101, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9b764fd6.py", line 67, in run_trial
    D_f = communication_discrepancy(xor_function, n)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9b764fd6.py", line 52, in communication
    D += abs(f[(x, y)] - f[(x + 1) % (2**n)][(y + 1) % (2**n)])
           ~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

KeyError: 1

Judge reasoning

Test crashed with KeyError during computation of communication discrepancy, preventing validation of conjecture | next: Fix the KeyError in communication_discrepancy function by ensuring all (x,y) pairs are properly initialized in the XOR-function's domain

Hypergraph Matching Complexity for Monotone DNF

- **Verdict:** INCONCLUSIVE
- **Bridge:** Hypergraph Theory × Monotone DNF Formulas
- **Recorded:** 2026-05-11 00:37 UTC
- **Entry ID:** 5de245e2396b

Statement

For any monotone DNF formula with m terms over n variables, the maximum matching size μ satisfies: (i) $\mu \leq \log n + 2\sqrt{m}$ for $m \leq n^2$, (ii) $\mu \geq n/2$ for k -CLIQUE DNFs with $k \geq 2\sqrt{n}$.

Rationale

Hypergraph matching captures structural limitations of DNFs; its growth rate could expose inherent complexity gaps between polynomial-size DNFs and k -CLIQUE's exponential requirements. The conjecture links hypergraph expansion properties to circuit size lower bounds.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.11s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5734af53.py", line 94, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5734af53.py", line 68, in run_trial
    matching_size = len(greedy_matching(hypergraph))
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5734af53.py", line 50, in greedy_matching
    matching.append(list(hypergraph_copy[max_edge_index]))
                     ~~~~~^~~~~~
```

TypeError: list indices must be integers or slices, not NoneType

Judge reasoning

Test crashed due to TypeError, preventing result collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Debug the hypergraph indexing error in test_5734af53.py and re-run experiments

Additive Energy Threshold for ACC⁰ Circuit Size in CNF Formulas

- **Verdict:** INCONCLUSIVE
- **Bridge:** Additive Combinatorics × ACC⁰ Circuit Size for CNF Formulas
- **Recorded:** 2026-05-11 00:43 UTC
- **Entry ID:** 0553f6013ae8

Statement

For any CNF formula Φ with n variables, if the additive energy of the set of satisfying assignments S is $\Omega(n^3)$, then Φ cannot be computed by an ACC^0 circuit of size $o(n^2)$.

Rationale

Additive energy measures the structure of the solution set; ACC^0 circuits have limited capacity to capture such complex structures, leading to a lower bound on circuit size.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_89dca52f.py", line 83, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_89dca52f.py", line 63, in run_trial
    energy = add_energy(cnf, assignment)
             ^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_89dca52f.py", line 49, in add_energy
    for x, y, z, w in combinations(range(n), 4):
                                   ^
```

NameError: name 'n' is not defined

Judge reasoning

Test crashed due to undefined variable 'n' in the add_energy function, preventing data collection. |
next: Fix the NameError in test_89dca52f.py by defining 'n' before using it in combinations

SOS Refutation Size Bounded by Curve Genus of CSP Constraints

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic Geometry (Curves over Finite Fields) × SOS Refutation Size for CSP Instances
- **Recorded:** 2026-05-11 00:50 UTC
- **Entry ID:** 66fe027cc0e4

Statement

For a CSP instance over $\text{GF}(2^n)$ with constraints defining a smooth projective curve C of genus g , the SOS refutation size is at most $2^{O(g \log n)}$. Conversely, if the curve has genus $g \geq \log n$, the SOS refutation size exceeds $2^{\Omega(g)}$.

Rationale

The genus captures the topological complexity of the solution space, influencing the number of moments required for SOS to capture contradictions. High-genus curves may encode inherently complex interactions between constraints, necessitating exponential refutation depth.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s

```
9465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
RESULT: SUPPORTED mean=12.896247699465729 std=0.0 support_fraction=1.0
```

Judge reasoning

Safety rail: critic_challenge | original judge: All 30 tested instances satisfy the conjecture with consistent metric values and no counterexamples found. | next: Test larger genus values and higher n to validate the exponential bounds

Tutte Polynomial Evaluation Bounded by Monotone k-CLIQUE Circuit Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Tutte Polynomial Theory $\times$ Monotone Circuit Size for k-CLIQUE
- **Recorded:** 2026-05-11 01:05 UTC
- **Entry ID:** 954a5c8e8ffd

Statement

For any k -CLIQUE instance on n vertices, the Tutte polynomial evaluated at $(2, 1)$ satisfies $T(2,1) \geq \Omega(n^{\lfloor k/2 \rfloor})$ and $T(1,2) \geq \Omega(n^{\lfloor k/2 \rfloor})$, where $T(x,y)$ is the Tutte polynomial of the graphic matroid induced by the complete graph K_n .

Rationale

The Tutte polynomial encodes matroid structure, and its evaluation at $(2,1)$ and $(1,2)$ captures connectivity properties. For k -CLIQUE, these values may correlate with the exponential growth of monotone circuit size due to the combinatorial explosion in clique enumeration.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12ec44ec.py", line 99, in <module>
    result = run_trial(seed)
    ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12ec44ec.py", line 72, in run_trial
    T21, T12 = get_tutte_values(graph, n // 2 + 1)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12ec44ec.py", line 61, in get_tutte_val
    T21 = tutte_polynomial(graph)
    ^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12ec44ec.py", line 29, in tutte_polynom
    if (graph,) in memo:
    ^^^^^^^^^^^^^^^^^
```

TypeError: unhashable type: 'collections.defaultdict'

Judge reasoning

Test crashed due to unhashable defaultdict graph input | next: Fix graph representation to use hashable structures like frozenset for memoization

Young Tableau Count Exponential Gap in Disjointness Communication Matrices

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic Combinatorics (Standard Young Tableau Count) $\times$ Communication Complexity (Disjointness Problem)
- **Recorded:** 2026-05-11 01:20 UTC
- **Entry ID:** 2b7d955e959e

Statement

For a random disjointness instance with n variables, the number of standard Young tableaux of shape λ (where λ is the partition derived from the communication matrix) is exponentially larger than for the corresponding determinant matrix, implying a lower bound on the communication complexity.

Rationale

The exponential gap in tableau counts reflects the combinatorial structure of the communication matrix, which could be leveraged to derive lower bounds via representation-theoretic methods.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_525950f5.py", line 86, in <module>
```

```

    result = run_trial(seed)
    ^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_525950f5.py", line 66, in run_trial
    det_tableau_count = count_standard_young_tableaux(det_partition)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_525950f5.py", line 58, in count_standard_young_tableaux
    return hook_length_formula(partition)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_525950f5.py", line 26, in hook_length_formula
    for cell, size in enumerate(row):
    ^^^^^^^^^^^^^^^^^
TypeError: 'int' object is not iterable

```

Judge reasoning

Test crashed due to `TypeError`, preventing data collection. The crash indicates a code bug rather than a conjecture failure. | next: Fix the `hook_length_formula` implementation to handle partitions correctly

Fourier Coefficient Sum Inverse Proportional to Disjointness Discrepancy

- **Verdict:** INCONCLUSIVE
- **Bridge:** Fourier Analysis of Boolean Functions × Communication Complexity of DISJOINTNESS
- **Recorded:** 2026-05-11 01:40 UTC
- **Entry ID:** 982d20a21495

Statement

For every n -variable DISJOINTNESS instance, the sum of absolute Fourier coefficients of its characteristic function satisfies $\sum |\hat{f}(S)| = \Theta(1/\Delta)$, where Δ is the discrepancy of the communication matrix. Equality holds when the instance is a balanced complete bipartite graph.

Rationale

Fourier coefficients capture correlation with parity functions, while discrepancy measures communication complexity. Their inverse proportionality suggests hidden structural alignment between spectral properties and combinatorial communication lower bounds.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.06s

```

ed": 463, "metric_name": "Fourier Coefficient Sum", "metric_value": 466.0, "instances_tested": 40, "config": {"seed": 503, "metric_name": "Fourier Coefficient Sum", "metric_value": 548.0, "instances_tested": 40},
TRIAL: {"seed": 503, "metric_name": "Fourier Coefficient Sum", "metric_value": 548.0, "instances_tested": 40},
TRIAL: {"seed": 547, "metric_name": "Fourier Coefficient Sum", "metric_value": 438.0, "instances_tested": 40},
TRIAL: {"seed": 593, "metric_name": "Fourier Coefficient Sum", "metric_value": 476.0, "instances_tested": 40},
TRIAL: {"seed": 631, "metric_name": "Fourier Coefficient Sum", "metric_value": 454.0, "instances_tested": 40},
TRIAL: {"seed": 677, "metric_name": "Fourier Coefficient Sum", "metric_value": 504.0, "instances_tested": 40},
TRIAL: {"seed": 727, "metric_name": "Fourier Coefficient Sum", "metric_value": 564.0, "instances_tested": 40},
TRIAL: {"seed": 773, "metric_name": "Fourier Coefficient Sum", "metric_value": 440.0, "instances_tested": 40},

```

```

TRIAL: {"seed": 821, "metric_name": "Fourier Coefficient Sum", "metric_value": 632.0, "instances_tested": 1, "conjecture_holds": True}
TRIAL: {"seed": 877, "metric_name": "Fourier Coefficient Sum", "metric_value": 504.0, "instances_tested": 1, "conjecture_holds": True}
TRIAL: {"seed": 929, "metric_name": "Fourier Coefficient Sum", "metric_value": 616.0, "instances_tested": 1, "conjecture_holds": True}
RESULT: FALSIFIED counterexample="mapping_undefined" first_failing_seed=11

```

Judge reasoning

Safety rail: critic_challenge_falsified | original: Test results show conjecture_holds is False for multiple seeds with counterexample 'mapping_undefined'. The presence of any counterexample meets the f | next: Investigate the 'mapping_undefined' counterexample to determine specific DISJOINT-NESS instances violating the conjecture

SOS Refutation Degree Bounded by Clause Density for Random 3-SAT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry $\times$ 3-SAT Instances
- **Recorded:** 2026-05-11 01:51 UTC
- **Entry ID:** 52c162cccd9b

Statement

For a random 3-SAT instance with n variables and m clauses, the minimal SOS degree required to refute it is $\Theta(\sqrt{m})$.

Rationale

The SOS hierarchy's refutation degree captures the algebraic complexity of constraint systems. By linking it to clause density, we expose geometric structure in SAT solving that could yield new lower bounds for refutation algorithms.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

```

569495584, got inf"}
TRIAL: {"seed": 593, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjecture_holds": False}
TRIAL: {"seed": 631, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjecture_holds": False}
TRIAL: {"seed": 677, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjecture_holds": False}
TRIAL: {"seed": 727, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjecture_holds": False}
TRIAL: {"seed": 773, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjecture_holds": False}
TRIAL: {"seed": 821, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjecture_holds": False}
TRIAL: {"seed": 877, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjecture_holds": False}
TRIAL: {"seed": 929, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjecture_holds": False}

```

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0e7c05f5.py", line 93, in <module>
    first_failing_seed = next(r['seed'] for r in results if not r['conjecture_holds'])
                        ~~~~~~
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0e7c05f5.py", line 93, in <genexpr>
    first_failing_seed = next(r['seed'] for r in results if not r['conjecture_holds'])

```


Judge reasoning

parse_fail | next: NONE

Schatten p-Norm Lower Bound on Disjointness Communication Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Noncommutative L^p Geometry $\times$ Communication Complexity of DISJOINTNESS
- **Recorded:** 2026-05-11 01:59 UTC
- **Entry ID:** aa045d952985

Statement

For any $n \geq 1$, the Schatten p-norm of the communication matrix for the DISJOINTNESS problem on n-bit inputs is $\Omega(n)$, where p is the dual exponent satisfying $1/p + 1/q = 1$ for the optimal lower bound.

Rationale

The Schatten p-norm captures the operator structure of the communication matrix, which is tightly linked to the required information-theoretic resources in multi-party communication. For DISJOINTNESS, the norm's growth with n reflects the inherent difficulty of coordinating disjointness checks.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.14s

```
nstances_tested': 30, 'conjecture_holds': False, 'counterexample': 'Schatten p-  
norm does not scale linearly with n'}  
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 10.011753727979853, 'instances_tested': 30,  
norm does not scale linearly with n'}  
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.731746236597688, 'instances_tested': 30,  
norm does not scale linearly with n'}  
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 10.011658675891052, 'instances_tested': 30,  
norm does not scale linearly with n'}  
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.672856354946585, 'instances_tested': 30,  
norm does not scale linearly with n'}  
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 8.734210406193428, 'instances_tested': 30,  
norm does not scale linearly with n'}  
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.586306087041859, 'instances_tested': 30,  
norm does not scale linearly with n'}  
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.752608783128863, 'instances_tested': 30,  
norm does not scale linearly with n'}  
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.97251009486534, 'instances_tested': 30,  
norm does not scale linearly with n'}  
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.914781880120312, 'instances_tested': 30,  
norm does not scale linearly with n'}
```

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_be69cc65.p

Judge reasoning

Test crashed with ZeroDivisionError, preventing reliable results. Pre-registered support condition cannot be evaluated without valid data. | next: Fix the test to handle empty results and re-run with sufficient sample size

Matroid Rank Lower Bound for Monotone k-CLIQUE

- **Verdict:** INCONCLUSIVE
- **Bridge:** Matroid Theory $\times$ Monotone Circuit Size for k-CLIQUE
- **Recorded:** 2026-05-11 03:08 UTC
- **Entry ID:** cf5188cf0224

Statement

For a monotone DNF formula φ , let M_φ be the matroid whose ground set is clauses of φ and rank function $r(S) = \max\{|S'| : S' \subseteq S, S' \text{ is a collection of pairwise disjoint clauses}\}$. Then $r(M_\varphi) = O(\log n)$ for all φ with circuit size $\leq n^c$, but $r(M_\varphi) = \Omega(n)$ for all k-CLIQUE instances with $k \geq 3$.

Rationale

Matroid rank captures combinatorial independence, which may mirror the structural constraints of monotone circuits. High rank for k-CLIQUE suggests inherent complexity, while low rank for general formulas hints at decomposability, offering a novel angle to circuit lower bounds.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.02s

TIMEOUT after 240s

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with extended computation limits

Symmetric Power Decomposition Complexity Gap for 3-SAT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Schur-Weyl Duality $\times$ 3-SAT Instances
- **Recorded:** 2026-05-11 03:39 UTC
- **Entry ID:** 9f467f497a7b

Statement

For a 3-SAT instance with n variables, the number of irreducible components in the symmetric power decomposition of its clause-variable incidence matrix is $\Omega(2^n)$ larger than for the corresponding determinant's decomposition.

Rationale

Schur-Weyl duality decomposes tensor powers into irreducible representations; permanent's symmetric nature yields exponentially more components than determinant's alternating structure, exposing algebraic barriers to polynomial identity testing.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.39s

79426043297591592576425399311708450748681513839438964967856509413642281551708967083891786846523124
TRIAL: {'metric_name': 'Symmetric Power Decomposition Complexity Gap', 'metric_value': 0, 'instances_

Judge reasoning

Test crashed before producing valid data, rendering the metric value unreliable. | next: Re-run test with increased time limits and verify code stability

Specht Module Decomposition Exponential Gap Between Permanent and Determinant Polynomials

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic combinatorics (Specht modules, symmetric functions) $\times$ SOS refutation degree for polynomial CSP instances
- **Recorded:** 2026-05-11 04:26 UTC
- **Entry ID:** 5afbe3121f1c

Statement

For random n -variable polynomial CSP instances derived from the permanent and determinant, the SOS refutation degree of the permanent instance is exponentially larger than that of the determinant instance for all $n \geq 2$.

Rationale

The exponential growth in the number of Specht modules in the permanent's decomposition suggests higher structural complexity, which could correlate with increased SOS refutation requirements.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.05s

TIMEOUT after 240s

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and higher resource allocation to complete execution

Persistent Homology Gap in Read-Twice BPs for IP_2

- **Verdict:** INCONCLUSIVE
- **Bridge:** Persistent Homology (Topological Data Analysis) × Read-Twice Branching Programs
- **Recorded:** 2026-05-11 04:32 UTC
- **Entry ID:** f6f1e06ee5ff

Statement

For read-twice BP P computing IP_2 , the maximum persistence of 1-dimensional homology classes in the simplicial complex built from P 's state transition graph satisfies $\text{persistence}(P) \geq \Omega(n)$. For general read-twice BPs computing any function f , $\text{persistence}(P) \leq O(\log \text{size}(P))$

Rationale

Persistent homology captures topological robustness of state transition structures. IP_2 's inherent symmetry creates persistent holes in its BP structure, while general functions' transitions collapse faster. This links algebraic topology to BP size complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.04s

```
dom': False, 'counterexample': 'IP_2 BP failed: persistence(P) < Ω(n)\nRandom function BP failed: pers
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 3, 'metric_value_random': 3, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 3, 'metric_value_random': 3, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 4, 'metric_value_random': 4, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 4, 'metric_value_random': 4, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 9, 'metric_value_random': 9, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 9, 'metric_value_random': 9, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 19, 'metric_value_random': 19, 'instances_
```

Judge reasoning

Insufficient instances_tested=2 to validate asymptotic bounds. Counterexamples may stem from small- n behavior rather than general complexity. | next: Test with larger n (e.g., $n \geq 1000$) to observe asymptotic scaling of persistence metrics

Free Entropy Gap in Read-Twice BP Transition Matrices

- **Verdict:** INCONCLUSIVE
- **Bridge:** Free Probability Theory × Read-Twice Branching Programs
- **Recorded:** 2026-05-11 04:50 UTC
- **Entry ID:** 9858410e3069

Statement

Let P be a read-twice BP computing a function $f: \{0,1\}^n \rightarrow \{0,1\}$. Define M_P as the $n \times n$ matrix where $M_P[i][j] = (\text{number of paths from root to leaf } i \text{ that read bit } j \text{ twice})$. The free entropy $\phi(M_P)$ satisfies $\phi(M_P) \leq \log(\text{size}(P)) + O(\sqrt{n})$, but for IP_2 (inner product mod 2), $\phi(M_P) \geq \Omega(n \log n)$.

Rationale

Free entropy quantifies non-commutative randomness in matrix laws; read-twice BPs exhibit structured non-commutativity via repeated bit access. This gap could reveal inherent complexity in repeated access patterns, analogous to how Shannon entropy bounds communication complexity.

Novelty

- Judge: INCONCLUSIVE over 5 arXiv hits

Top hits consulted: - [1201.0716v2] Concavification of free entropy - [1510.05486v2] Two types of branching programs with bounded repetition that cannot efficiently compute monotone 3-CNFs - [1902.03873v2] Analogues of Entropy in Bi-Free Probability Theory: Non-Microstate - [1809.06500v3] Range entropy: A bridge between signal complexity and self-similarity - [1902.03874v3] Analogues of Entropy in Bi-Free Probability Theory: Microstates

Empirical Test

- exit code: 1, elapsed: 1.25s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5342c042.py", line 90, in <module>
    result = run_trial(seed)
             ~~~~~^~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5342c042.py", line 64, in run_trial
    R = r_transform(M_P)
       ~~~~~^~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5342c042.py", line 44, in r_transform
    if M[i][j] > 0:
       ~~~~^
```

IndexError: list index out of range

Judge reasoning

Test crashed with IndexError before producing data; cannot assess support fraction or counterexamples | next: Debug `r_transform` implementation to handle matrix indexing properly